



# Sustainability Capstone: Decarbonising Travel

**Accelerating Net Zero Through Process Intelligence and Data-Driven Decision Making**

Speaker name: **Deneb Kumar Jena**  
Institute: GGSIP University, Delhi  
Branch & Year of Study: **2008**  
Email ID: **deneb.jena@gmail.com**

# Executive Summary – Towards Net Zero Business Travel by 2030



| KPI                                    | Value                       |
|----------------------------------------|-----------------------------|
| Total Travel CO <sub>2</sub> Emissions | 178.1M kg CO <sub>2</sub> e |
| Total Trips                            | 65,289                      |
| High-Carbon Trips                      | 16,326 (25%)                |
| Average CO <sub>2</sub> per Trip       | 2,728 kg CO <sub>2</sub> e  |
| Average CO <sub>2</sub> per Employee   | 3209 kg CO <sub>2</sub> e   |

## Top 3 Insights

- 84% of travel emissions originate from flights, making aviation the primary decarbonization opportunity.
- Customer Visits (43%) and the Sales business unit (26%) are the largest contributors to travel emissions.
- 373 unique travel process variants identified; however, 87% of trips follow just 34 variants, indicating strong opportunities for workflow standardization.

| Estimated Impact                                                         |                                           |
|--------------------------------------------------------------------------|-------------------------------------------|
| Potential reduction of emissions (≈26% of the current travel footprint). | <b>46</b><br>million kg CO <sub>2</sub> e |
| Estimated travel cost savings                                            | <b>\$14</b><br>million                    |

## Top 3 Recommendations

- Restrict Business & First-Class Flight travel and promote rail alternatives on feasible European region.
- Adopt a **Virtual-First Policy** for internal meetings and introductory customer engagements to reduce unnecessary travel.
- Deploy **AI-powered Carbon-Intelligence Travel App** with carbon budgets, sustainable transport recommendations (carbon calculator), and real-time policy compliance using Celonis Process Mining.

# Emissions Profile and Benchmarking



Celonis sustainability report indicates Scope 3 carbon emissions stems from employee business travel **contributing 47.6% towards total emissions**[1].

## Transport Mode Footprint

**Flights account for approximately 84% of total travel emissions**, making air travel the primary decarbonization opportunity. Rail and Ground Transit ( $\approx 15\%$ )

**Trains and EVs account for 1.5% of emissions** offering promising savings in costs and carbon emissions for net zero objective.

**Petrol/Diesel Cars account for 14% of emissions.** Replacing them with EVs can have significant impact.

## Regional Footprint

Regional analysis shows that **Australia, Brazil, China** are the largest departure emission sources.

**Route Analysis** shows that **Extra Long Haul & Long Haul Routes** account for **75%** of the Travel Emissions with Business & Economy Flights contributing 67% alone.

All routes with a **100% High Carbon Rate (AU→DE, BR→JP, CN→TR)** belong to the Long/Extra Long-Haul route & account for 25% of Emissions.

## Industry Benchmark

Scope 3 emissions dominate the carbon footprint of SaaS and technology companies, often contributing to more than 90% of total emissions, making business travel an important decarbonization focus.[3][4]

Celonis travel emissions are consistent with SaaS companies where employee travel is one of the largest Scope 3 categories.

Celonis aligns with CDP benchmarks by targeting per-FTE intensity reductions (51.6%) to ensure the company lowers its environmental impact even as the business scales.[2]

# Hotspot Analysis

## Business Unit Hotspots

**Sales** is the largest contributor, generating **47.1 million kg CO<sub>2</sub>e (26.4%)** of total travel emissions, primarily driven by extensive customer-facing travel.

**Marketing (14.8%)** and **Value Engineering (7.8%)** are the next highest contributors, indicating that commercial and pre-sales functions account for the majority of travel-related emissions.

## Travel Purpose Hotspots

**Customer Visits** account for **76.9 million kg CO<sub>2</sub>e (43.2%)**, making them the single largest source of travel emissions.

**Conferences and Exhibitions** contribute **35.4%** of total emissions, highlighting external engagement activities as another significant carbon hotspot. The above 2 purposes are the highest-priority areas for emission reduction.

## High-Impact Travel Routes

- **Australia → Germany (Extra Long Haul Route)** is the largest carbon hotspot, generating **37.9 million kg CO<sub>2</sub>e** across **5,366 trips**, with **100% classified as high-carbon**.
- **Australia → United States (Extra Long Haul Route)** contributes **28.2 million kg CO<sub>2</sub>e**, making it the second-largest hotspot route despite a lower high-carbon rate (**47%**).
- **Brazil → United States (15.5 million kg CO<sub>2</sub>e)** and **Australia → Singapore (13.2 million kg CO<sub>2</sub>e)** also contribute substantially due to their high travel volumes.
- Although **Brazil → Japan** contributes a smaller total volume (**5.5 million kg CO<sub>2</sub>e**), it records the **highest average emissions per trip (7,693 kg CO<sub>2</sub>e)** with a **100% high-carbon rate**, making it one of the least carbon-efficient routes.
- **China → United States** must not be ignored for its **highest average emissions per trip (7,606 kg CO<sub>2</sub>e)**.

# Inefficiencies Identified



## Process Inefficiencies

- 1. High Process Variation:** 373 unique travel process variants were identified, indicating inconsistent travel execution. While the **top 34 variants account for 87% of all trips**, the remaining **339 variants** have opportunities for workflow standardization.
- 2. Long Process Cycle Time:** The **average end-to-end travel lifecycle duration is 22.3 days**. Also, the travel process records emissions **after travel is completed**, but employees currently receive no carbon guidance during planning.
- 3. Heavy Dependence on Air Travel:** **Aviation account for approximately 84% of total travel emissions**. Departure and Return flights have same travel class (increases cost and emissions).
- 4. Emission Concentration:** Few business units (**Sales & Customer Visits**) and Few Routes (**AU → DE, AU → US, BR → US**) contribute to highest share of carbon footprint. Small policy changes can have significant impact.

## Data Inconsistencies

**Route infeasibility:** A number of train journeys and road journeys (e.g., China → India/USA/Aus/GB) appear operationally infeasible, suggesting synthetic routing assumptions within the dataset.

In some routes the average netcosts of business class trips is lower than average netcosts of economy class trips.

**Employee travel continuity:** Consecutive trips for the same employee (eg. M1009-M5146) sometimes originate from unrelated cities without recorded repositioning travel, indicating that employee location continuity is not explicitly modeled.

**Emission drivers:** Hotel category impacts emissions than duration alone, creating a need to record hotel category to reduce emissions by prioritizing low-carbon hotels.

# Low Carbon Strategic Recommendations (Quantified)

| Top Recommendations                                                                                                                        | Data-backed Justification                                                                                                                                                                                                | Estimated CO <sub>2</sub> Reduction                       | Estimated Cost Savings                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------|
| Restrict Business & First-Class Travel (VP level & above)                                                                                  | Business & First-Class flights are responsible for ≈80 million kg CO <sub>2</sub> e (45% of total travel emissions). Reducing premium travel by 20% with stricter approval policies would significantly lower emissions. | ≈16M kg CO <sub>2</sub> e                                 | ≈\$6.8M                                                  |
| Optimize High-Emission International Routes (AU→DE, AU→US, BR→US, AU→SG, BR→JP)[1]<br>And Extra Long Haul, Long Haul and Short Haul Routes | These 5 routes alone generate ≈100M kg CO <sub>2</sub> e (56% of total travel emissions). Conservative scenario[1] can reduce emissions by total 11%.                                                                    | ≈11.6M kg CO <sub>2</sub> e +<br>≈8M kg CO <sub>2</sub> e | ≈\$0.5M                                                  |
| Adopt a Virtual-First Policy for introductory client meetings and internal meetings                                                        | Customer Visits generate 76.9M kg CO <sub>2</sub> e (43%) of total emissions. Replacing just 10% of these trips with virtual meetings would reduce unnecessary travel while maintaining client engagement.               | ≈7.7M kg CO <sub>2</sub> e                                | ≈\$6M                                                    |
| Replace Return Flights with Rail on Feasible European Corridors (e.g., DE→FR, DE→NL, DE→IT, DE→ES and domestic Germany)                    | Rail-connected European routes currently generate ≈1.67M kg CO <sub>2</sub> e from Business & First-Class flights alone. Replacing these flights with rail returns is a quick-win solution.[2]                           | ≈3.3M kg CO <sub>2</sub> e                                | ≈\$0.5M<br>(No return airfare and airport-related costs) |
| Deploy AI-powered Carbon-Intelligence Travel app using Celonis                                                                             | Integrate carbon budgets, approval workflows, low emission recommendations, carbon calculators and policy checks into the booking process to prevent high-carbon travel before it is booked.                             | Continuous long-term reduction                            | Reduced travel and administrative costs                  |

**Estimated Impact:** 46 million kg CO<sub>2</sub>e Savings & ≈ \$14 million Cost Savings  
Reduced dependence on premium flights & unnecessary travel. Improved policy compliance through AI and Process Mining.

# Strategic Integration Roadmap

## Build the Foundation

- Introduce Virtual-First Policy for internal meetings and introductory client engagements.
- Restrict Business & First-Class travel through approval workflows.
- Launch carbon-aware travel dashboard in Celonis.

2026

## Optimize Travel Decisions

- Replace rail-feasible European flights with train.
- Optimize high-emission routes (AU→DE, AU→US, BR→US, AU→SG).

2027

## Intelligent Net-Zero Travel

- Establish regional Solution centers.
- Real-time Process Mining to detect policy deviations.

2029

2028

2030

## Continuous Monitoring

- Continuous carbon monitoring by department and employee.
- Integrate travel data with corporate ESG reporting.

## Partner and Deploy

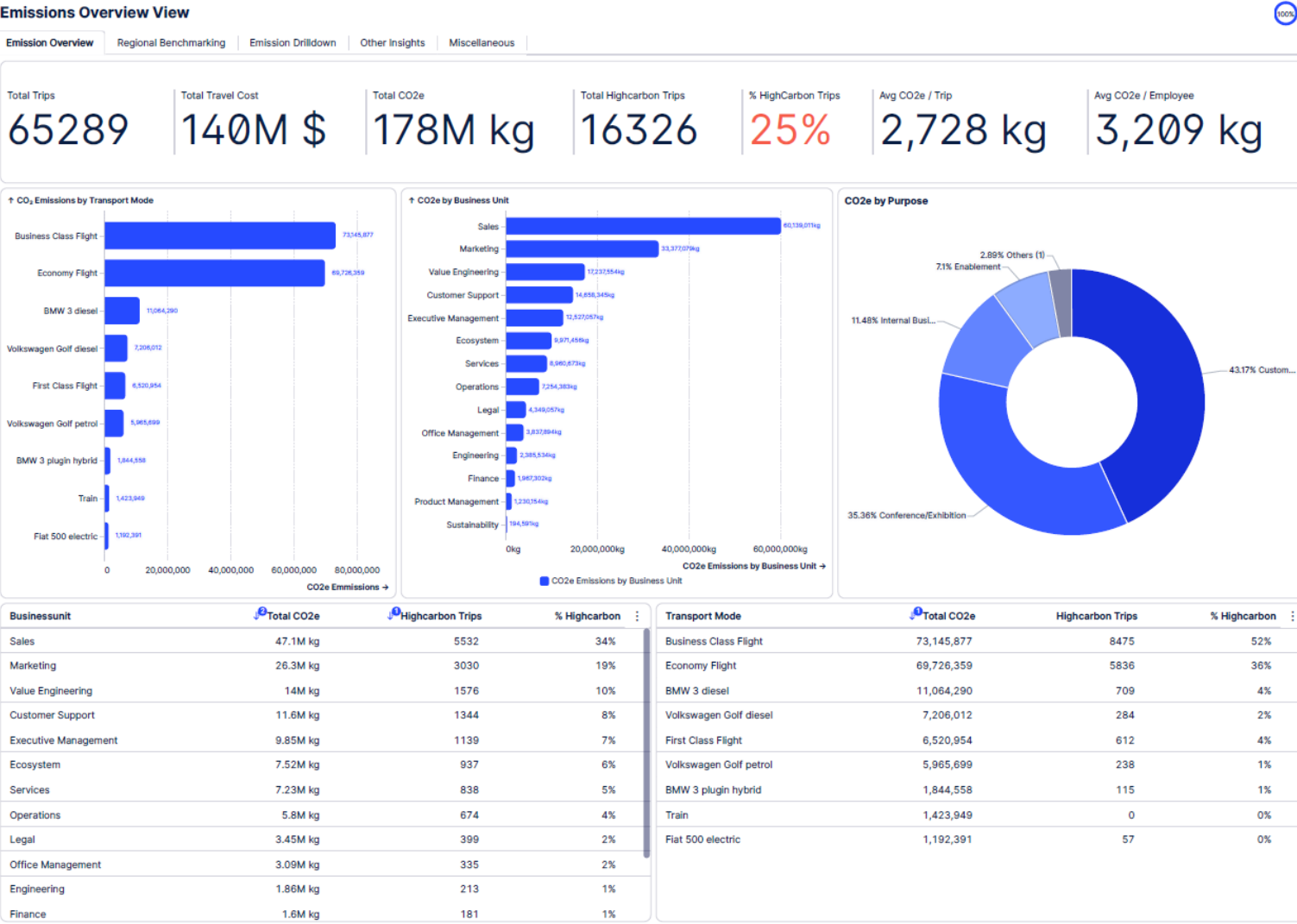
- Partner with airlines offering Sustainable Aviation Fuel (SAF).
- Deploy AI-powered Carbon-Intelligence Travel App for calculating trip emissions, recommendations, emission tracker and carbon budgets.



1. Establish regional pre-sales and remote support centers reduce long-haul travel and maintain customer responsiveness. Value engineers can be positioned near client location.
2. Talk to enterprise clients keen on reducing their carbon footprint. Virtual meetings and technical support can benefit both celonis and its clients in reducing their overall carbon footprint.
3. Business Class and First-Class Travel can be restricted to VP level or above.
4. Consolidate multiple meeting in a single trip, reduces cost and emissions significantly.
5. Set carbon budgets per employee and business units.
6. Educate employees on decarbonization. Implement an incentive program that recognizes employees who consistently reduce corporate travel costs and minimize carbon emissions.
7. Build AI-powered Carbon-Intelligence Travel App/Assistant that can calculate trip emissions, recommends rail over flights, checks weather disruptions, flags Business Class bookings, checks carbon budget, recommends nearby colleagues instead of flying another employee.
8. Partner with airlines offering bulk discounts, Sustainable Aviation Fuel (SAF) and low-emission routes for unavoidable long-haul travel.
9. Resolve data level inconsistencies like transport mode infeasibility, missing record on hotel category, missing employee travel sequence.
10. Invest in inhouse fleet of electric vehicles for local and short haul routes.



# Emission Overview



# Regional Benchmarking

## Route and Region Analysis

### Emissions Overview View

Emission Overview

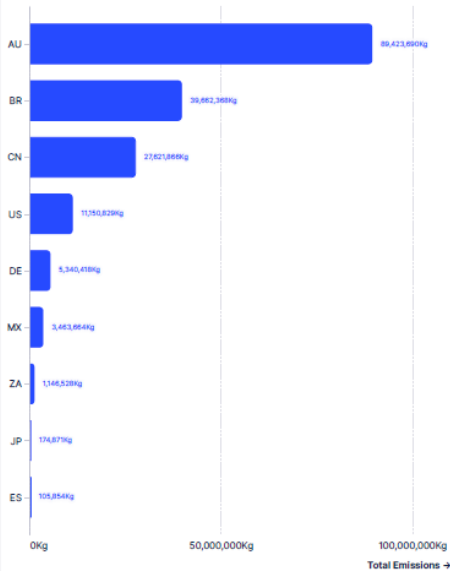
Regional Benchmarking

Emission Drilldown

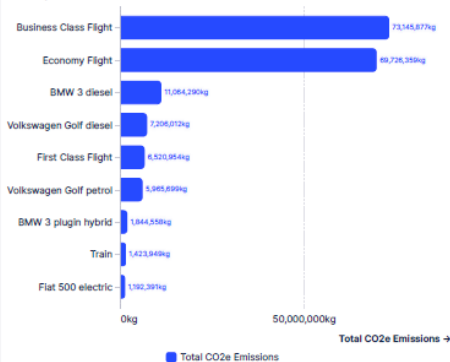
Other Insights

Miscellaneous

#### ↑ Total Emissions Departure Country



#### ↑ Transport Mode



#### Route Analysis

| Trip Route | Haul            | Total Trips | HighCarbon Trips | HighCarbon Rate | AVG CO2e / Route | Total CO2e / Route | Total Costs |
|------------|-----------------|-------------|------------------|-----------------|------------------|--------------------|-------------|
| AU → DE    | Extra Long Haul | 5366        | 5364             | 100%            | 7,066 kg         | 37,918,790 kg      | 11463750 \$ |
| AU → US    | Extra Long Haul | 5538        | 2621             | 47%             | 5,088 kg         | 28,177,466 kg      | 11980650 \$ |
| BR → US    | Long Haul       | 5716        | 1240             | 22%             | 2,708 kg         | 15,476,688 kg      | 12323400 \$ |
| AU → SG    | Medium Haul     | 4862        | 1188             | 24%             | 2,714 kg         | 13,195,950 kg      | 10519350 \$ |
| BR → ZA    | Long Haul       | 4079        | 894              | 22%             | 3,150 kg         | 12,850,020 kg      | 8713500 \$  |
| CN → RU    | Long Haul       | 3335        | 1101             | 33%             | 2,818 kg         | 9,399,603 kg       | 7110600 \$  |
| CN → IN    | Medium Haul     | 3634        | 202              | 6%              | 2,123 kg         | 7,713,743 kg       | 7773150 \$  |
| AU → CN    | Long Haul       | 1374        | 637              | 46%             | 4,403 kg         | 6,049,359 kg       | 2949900 \$  |
| BR → JP    | Extra Long Haul | 715         | 715              | 100%            | 7,693 kg         | 5,500,826 kg       | 1523100 \$  |
| US → MX    | Medium Haul     | 3231        | 0                | 0%              | 1,629 kg         | 5,264,671 kg       | 6943350 \$  |
| CN → AU    | Long Haul       | 838         | 417              | 50%             | 4,522 kg         | 3,789,645 kg       | 1792200 \$  |
| AU → AU    | Short Haul      | 3499        | 0                | 0%              | 1,061 kg         | 3,711,843 kg       | 7594200 \$  |
| MX → US    | Medium Haul     | 2016        | 0                | 0%              | 1,718 kg         | 3,463,664 kg       | 4343850 \$  |
| BR → PT    | Long Haul       | 939         | 215              | 23%             | 3,325 kg         | 3,122,586 kg       | 2040150 \$  |
| DE → ES    | Short Haul      | 3175        | 0                | 0%              | 933 kg           | 2,961,052 kg       | 6892050 \$  |
| US → FR    | Long Haul       | 758         | 574              | 76%             | 3,748 kg         | 2,840,750 kg       | 1651650 \$  |
| CN → US    | Long Haul       | 372         | 287              | 77%             | 7,606 kg         | 2,829,542 kg       | 790200 \$   |
| CN → GB    | Long Haul       | 713         | 331              | 46%             | 3,851 kg         | 2,745,808 kg       | 1530300 \$  |
| BR → AR    | Medium Haul     | 3225        | 0                | 0%              | 841 kg           | 2,712,250 kg       | 6889500 \$  |
| ZA → ZA    | Short Haul      | 1411        | 0                | 0%              | 813 kg           | 1,146,528 kg       | 3048600 \$  |
| CN → TR    | Long Haul       | 238         | 238              | 100%            | 4,805 kg         | 1,143,524 kg       | 526650 \$   |
| US → DE    | Long Haul       | 263         | 190              | 72%             | 3,972 kg         | 1,044,762 kg       | 542250 \$   |
| US → JP    | Long Haul       | 238         | 112              | 47%             | 4,370 kg         | 1,040,176 kg       | 475950 \$   |
| US → CA    | Medium Haul     | 2999        | 0                | 0%              | 320 kg           | 960,470 kg         | 6393450 \$  |
| DE → FR    | Short Haul      | 1795        | 0                | 0%              | 447 kg           | 802,904 kg         | 3850650 \$  |
| DE → IT    | Short Haul      | 916         | 0                | 0%              | 625 kg           | 572,290 kg         | 1971900 \$  |
| DE → NL    | Short Haul      | 1410        | 0                | 0%              | 365 kg           | 515,280 kg         | 3070050 \$  |
| DE → GB    | Short Haul      | 1146        | 0                | 0%              | 375 kg           | 429,888 kg         | 2481600 \$  |
| AU → NZ    | Medium Haul     | 610         | 0                | 0%              | 607 kg           | 370,282 kg         | 1358100 \$  |
| JP → JP    | Short Haul      | 273         | 0                | 0%              | 641 kg           | 174,871 kg         | 575700 \$   |
| ES → PT    | Other           | 288         | 0                | 0%              | 368 kg           | 105,854 kg         | 637050 \$   |
| DE → DE    | Other           | 317         | 0                | 0%              | 186 kg           | 59,004 kg          | 722400 \$   |

# Emission Drilldown

## Variants & Processes

### Emissions Overview View

Emission Overview | Regional Benchmarking | **Emission Drilldown** | Other Insights | Miscellaneous

34 process variants covers 87% of the cases

Process Variants

373

Avg Events / Trip

10

Avg Process Duration

22.31 Days

#### Variant Explorer

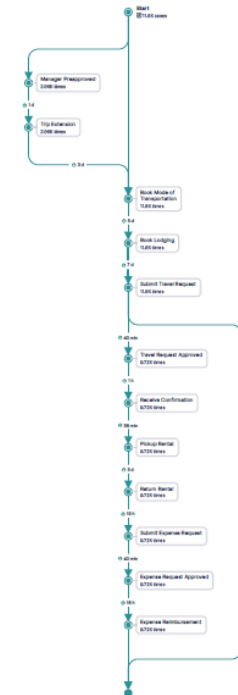
34 18% covered

#### Public Trip Event Log Eventname

18% cases covered  
2 of 373 variants  
11.8K of 65.3K cases

Apply Filter

| <input type="checkbox"/>            | Variant | Count | Coverage | Avg TPT |
|-------------------------------------|---------|-------|----------|---------|
| <input type="checkbox"/>            | #1      | 22.8K | 35%      | 25 d    |
| <input checked="" type="checkbox"/> | #2      | 9.72K | 15%      | 19 d    |
| <input checked="" type="checkbox"/> | #3      | 2.06K | 3%       | 5 d     |
| <input type="checkbox"/>            | #4      | 1.65K | 3%       | 20 d    |
| <input type="checkbox"/>            | #5      | 1.27K | 2%       | 25 d    |
| <input type="checkbox"/>            | #6      | 1.19K | 2%       | 25 d    |
| <input type="checkbox"/>            | #7      | 1.17K | 2%       | 25 d    |
| <input type="checkbox"/>            | #8      | 1.07K | 2%       | 28 d    |
| <input type="checkbox"/>            | #9      | 1.03K | 2%       | 28 d    |
| <input type="checkbox"/>            | #10     | 1.02K | 2%       | 28 d    |
| <input type="checkbox"/>            | Others  | 22.3K | 34%      | 21 d    |



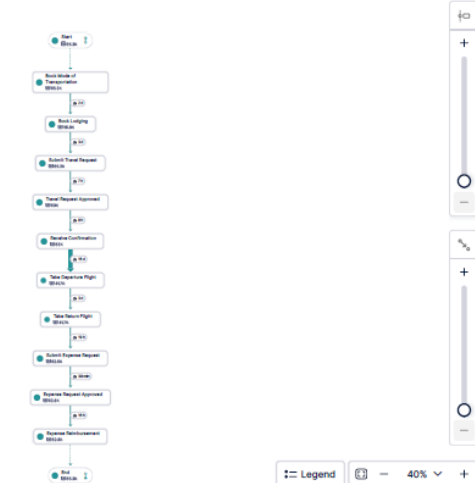
Legend 46% +

#### Variant Analysis

| VARIANT               | COUNT | Avg CO2e / Variant | Trip IDs               | Avg Cost | Avg TPT |
|-----------------------|-------|--------------------|------------------------|----------|---------|
| Manager Preapprov...  | 1     | 13,606 kg          | 74013                  | 1,500 \$ | 7 Days  |
| Book Mode of Trans... | 1     | 13,593 kg          | 24264                  | 1,950 \$ | 7 Days  |
| Submit Travel Requ... | 1     | 13,580 kg          | 53664                  | 1,500 \$ | 6 Days  |
| Book Mode of Trans... | 2     | 12,660 kg          | 75291, 85936           | 2,325 \$ | 10 Days |
| Book Mode of Trans... | 1     | 9,925 kg           | 30167                  | 300 \$   | 9 Days  |
| Book Lodging, Book... | 2     | 9,209 kg           | 9444, 31360            | 1,050 \$ | 12 Days |
| Manager Preapprov...  | 1     | 8,957 kg           | 37960                  | 3,600 \$ | 9 Days  |
| Book Mode of Trans... | 2     | 7,820 kg           | 16788, 83797           | 3,300 \$ | 6 Days  |
| Book Mode of Trans... | 1     | 7,470 kg           | 79590                  | 900 \$   | 12 Days |
| Submit Travel Requ... | 1     | 7,205 kg           | 23118                  | 2,850 \$ | 7 Days  |
| Book Mode of Trans... | 3     | 6,573 kg           | 2119, 7373, 33070      | 2,550 \$ | 9 Days  |
| Book Mode of Trans... | 4     | 6,461 kg           | 12820, 29062, 3263...  | 2,438 \$ | 9 Days  |
| Manager Preapprov...  | 25    | 6,429 kg           | 2231, 4586, 4656, 5... | 2,220 \$ | 7 Days  |
| Book Lodging, Book... | 1     | 6,266 kg           | 72302                  | 1,200 \$ | 6 Days  |
| Submit Travel Requ... | 1     | 6,214 kg           | 34002                  | 900 \$   | 5 Days  |
| Book Mode of Trans... | 27    | 5,976 kg           | 7568, 9378, 9806, 1... | 2,628 \$ | 12 Days |
| Book Mode of Trans... | 8     | 5,808 kg           | 7972, 29943, 32521,... | 2,475 \$ | 16 Days |
| Book Mode of Trans... | 5     | 5,759 kg           | 16573, 24806, 25901... | 2,160 \$ | 11 Days |
| Book Mode of Trans... | 2     | 5,602 kg           | 18188, 38457, 58127    | 2,250 \$ | 9 Days  |

#### Process Explorer

34



Legend 40% +



**THANK YOU**